UNIVERSITY OF BOHOL

College of Engineering, Technology, Architecture

And Fine Arts

Ma. Clara Street, Tagbilaran City, 6300 Bohol

"Transformer Rental Management System"

Project Member:

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BSCompE – 1

Supervisor/Adviser:

Engr. Liduvina Lagon Namocatcat

Introduction

The **Transformer Rental Management System (TRRMS)** is an automated system designed to manage the rental, return, and maintenance of transformers. The system ensures smooth stock management, accurate transaction logging, and secure user authentication for both cashiers and administrators. It addresses common challenges in manual systems, such as inventory errors, delayed processing, and lack of transparency, by providing real-time stock updates and transaction records.

Project Scope and Limitation

Scope:

1. **Transformer Rentals and Returns:**

   - Streamlines the rental and return process for transformers.
   - Facilitates rental duration management and cost computation.

2. **Stock Management:**

   - Monitors transformer stock levels, providing notifications when stock is low.
   - Updates stock dynamically as transactions occur.

3. **User Authentication:**

   - Ensures secure login for both cashiers and administrators.
   - Role-based access controls, with limited permissions for each user type.

4. **Transaction Logging and Receipt Generation:**

   - Automatically logs each transaction for audit purposes.
   - Generates detailed receipts that include transaction data, VAT breakdown, and grand total.

5. **Password Security:**

   - Masked password input ensures security during login.

Limitations:

1. **Console-Based Interface:**

- The system operates through a text-based console interface, requiring users to be familiar with command-line inputs.

2. **Manual Data Entry:**

- Initial stock and user data must be manually entered into the system files.

3. **Focus on Transformer Rentals:**

- The system is designed solely for managing transformer rentals, with no provisions for other equipment types.

4. **Limited Scalability:**

- The system relies on text file-based storage, limiting its ability to scale for large enterprises with large datasets.

Problem Requirements

Purpose:

To develop a comprehensive system that automates the transformer rental process, ensures efficient stock management, and provides accurate transaction records with secure user authentication.

Overall Description:

The system provides an intuitive, command-line interface where users can:

- Rent and return transformers.
- View and update stock levels.
- Generate transaction receipts, including VAT calculations.
- Admin and cashier roles with specific permissions for system access.

System References:

- **Screenshots:** Example screenshots from the testing phase show key system functions, such as stock check, transaction processing, and receipt generation.
- **References from Similar Systems:**
    - "Inventory and Asset Management System" by John Doe (Book Reference)
    - "Rental Systems: Automation and Optimization" by Jane Smith (Journal Article)

Analysis

Input Requirements:

- **User Credentials:** Username and password for login.
- **Transaction Details:** Customer name, venue, rental period, and transformer quantity.
- **Stock Data:** Current transformer stock (quantity, restock threshold, etc.).

Output Requirements:

- **Receipts:** Transaction details and cost breakdown for each rental.
- **Stock Alerts:** Notifications when stock falls below a predefined threshold.
- **Transaction Logs:** Detailed logs of each transaction for future audits.

Necessary Formulae and Their Descriptions:

1. **VAT Calculation:**
    - Formula: VAT = Total Cost * 0.12
    - Description: Calculates the 12% VAT to be added to the total rental cost.
2. **Grand Total Calculation:**
    - Formula: Grand Total = Total Cost + VAT
    - Description: Computes the grand total by adding VAT to the total cost.
3. **Stock Updates:**
    - For Rentals: Updated Quantity = Current Quantity - Number of Transformers
    - For Returns: Updated Quantity = Current Quantity + Number of Transformers

- Description: Updates the stock quantity based on rentals and returns.

Design

Files and Their Descriptions:

- **stock.txt:** Contains data about the available stock (transformer types, quantity, restock threshold).
- **users.txt:** Stores user credentials for authentication (username, password).
- **transactions.log:** Records all rental and return transactions, including customer details and cost.

User Interface Design:

The system uses a simple, menu-driven interface with the following options:

1. Check Stock
2. Start a New Transaction
3. Return Transformer
4. Logout
5. Switch User

Features of the Project:

- **Password-Masked Login:** Ensures password security during login.
- **Dynamic Stock Management:** Updates stock levels automatically after each transaction.
- **Transaction Logging:** All rental and return transactions are logged for future auditing.
- **VAT and Receipt Generation:** Automatically generates detailed receipts including VAT and grand total.

Security and Audit Considerations:

- **Input Masking:** Passwords are masked during entry to prevent unauthorized viewing.
- **Transaction Audits:** Every transaction is logged with relevant details to maintain an audit trail for accountability.

- **Role-Based Access:** Different user roles (admin and cashier) with restricted permissions to ensure proper access control.

Implementation

URL of the Saved Source Code:

https://drive.google.com/file/d/1mcEFcExawkH_LQPBcBjCLug7D4c-atue/view

Package Files and Their Descriptive Information:

- **Executable File:** transformer_rental (Compiled program for running the system).
- **Source Code File:** main.c (C source code file).
- **Data Files:**
  - stock.txt (Stock data).
  - users.txt (User credentials).
  - Transactions.log (Transaction logs).
  - Gainmoney.txt(rental and return money)

Function Declarations and Their Descriptions:

1. **void load_users**
   Loads user data from a file into an array for use in the program.
2. **void save_users**
   Saves user data from the program to a file for persistence.
3. **void load_stock**
   Loads stock information (like transformers) from a file into an array.
4. **void save_stock**
   Saves stock information from the program to a file for persistence.
5. **int check_stock**
   Checks if the required quantity of transformers is available for a given date.
6. **void update_stock**
   Updates stock information after a transaction, such as renting transformers.
7. **void return_transformer**
   Updates stock when transformers are returned.

8. **void initialize_stock**

   Initializes the stock file if it does not exist or is empty.
9. **char *trim_spaces**

   Removes extra spaces from the beginning and end of a string.
10. **void to_lower_case**

    Converts all characters in a string to lowercase.
11. **int is_valid_number**

    Checks if a string is a valid numeric value.
12. **int is_valid_date**

    Verifies if the provided date is in a valid format and range.
13. **int isBohecoMunicipality**

    Checks if a given municipality belongs to the Boheco region.
14. **int get_valid_choice**

    Ensures the user's choice is within a valid range of menu options.
15. **void generate_unique_filename**

    Creates a unique filename for saving or logging.
16. **int change_password**

    Allows a user to update their account password.
17. **int admin_login**

    Handles the process of logging in as an administrator.
18. **void display_menu**

    Displays the main program menu for user interaction.
19. **void display_stock**

    Shows the current inventory of stock items (e.g., transformers).
20. **void log_transaction**

    Records transaction details in a specified log file.
21. **void print_receipt**

    Prints a receipt with transaction details, costs, and approvals.
22. **void log_financial_transaction**

    (Commented out) Logs financial details such as payments or refunds.
23. **void print_file**

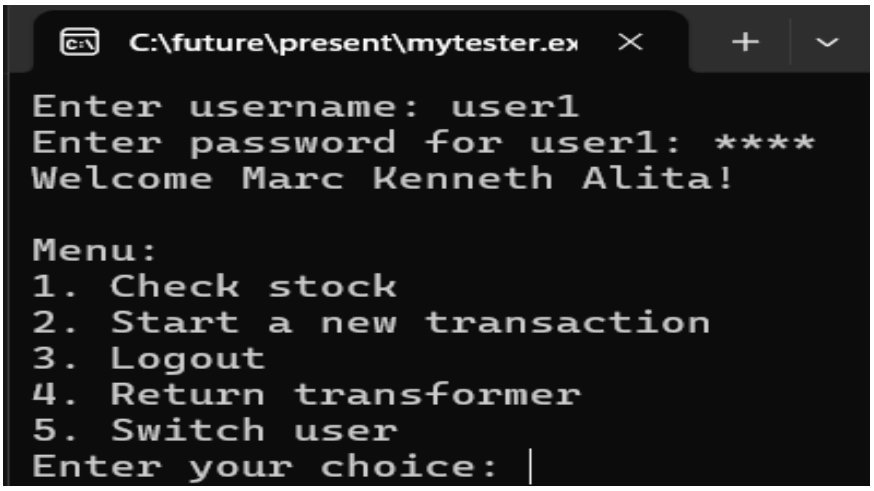
    Displays the content of a file in the console or another output.

Testing and Debugging

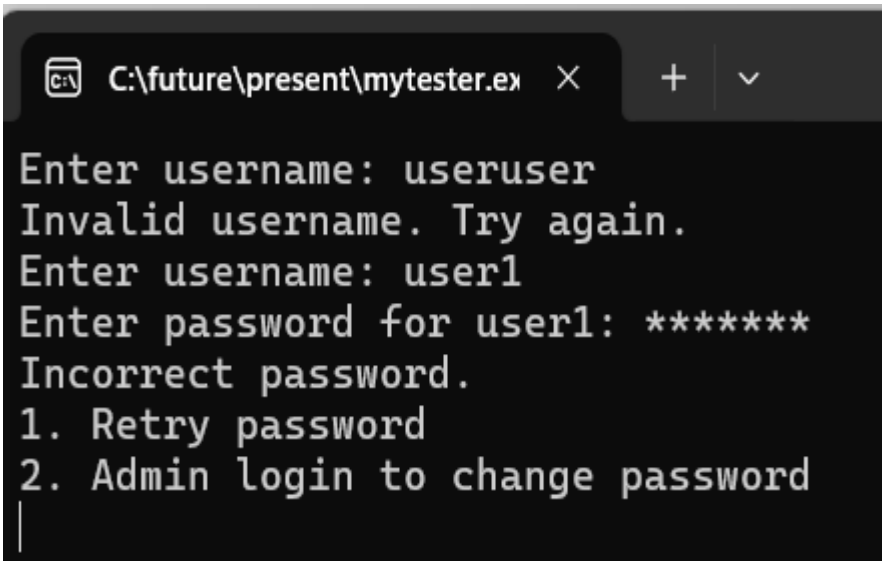
Screenshots of the Sample Run:

1. **Login Screen:**

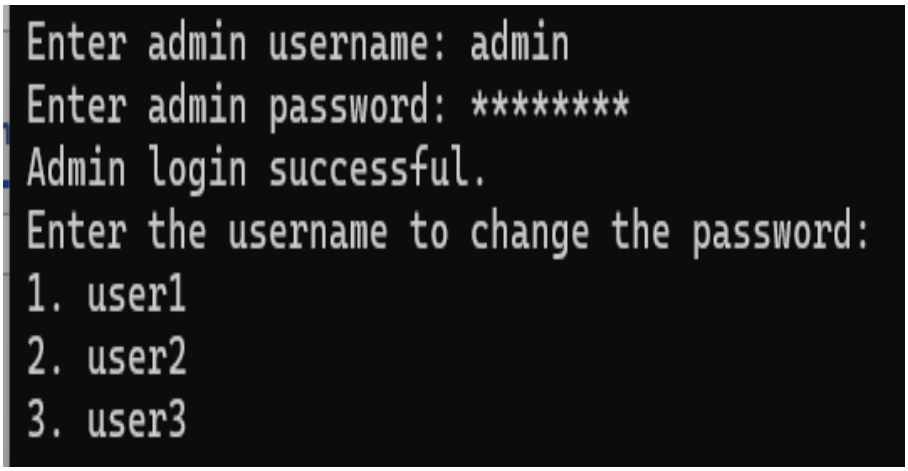
LOG IN USER SUCCESFUL:



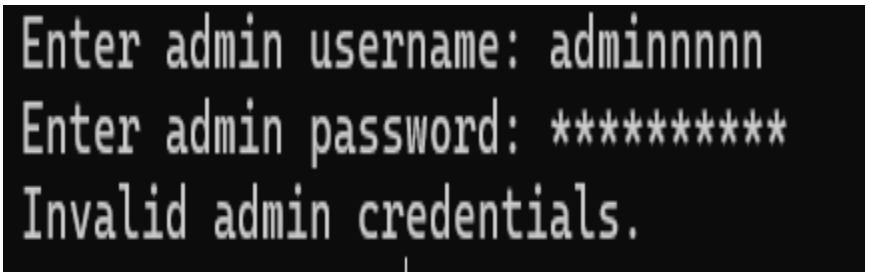
WRONG USER NAME AND PASSWORD:



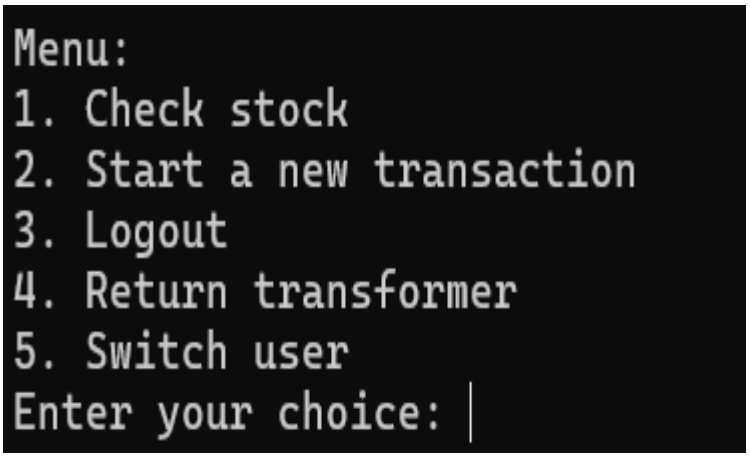
LOG IN ADMIN SUCCESFUL:



WRONG USER NAME AND PASSWORD FOR ADMIN:



2. **MENU:**



3. **Stock Check:**

```
5. Switch user
Enter your choice: 1
Stock on display:
Date: 2025-01-01, Quantity: 20
Date: 2025-01-02, Quantity: 20
Date: 2025-01-03, Quantity: 20
Date: 2025-01-04, Quantity: 20
Date: 2025-01-05, Quantity: 20
Date: 2025-01-06, Quantity: 20
Date: 2025-01-07, Quantity: 20
Date: 2025-01-08, Quantity: 20
Date: 2025-01-09, Quantity: 20
Date: 2025-01-10, Quantity: 20
Date: 2025-01-11, Quantity: 20
Date: 2025-01-12, Quantity: 20
Date: 2025-01-13, Quantity: 20
Date: 2025-01-14, Quantity: 20
Date: 2025-01-15, Quantity: 20
Date: 2025-01-16, Quantity: 20
Date: 2025-01-17, Quantity: 20
Date: 2025-01-18, Quantity: 20
Date: 2025-01-19, Quantity: 20
Date: 2025-01-20, Quantity: 20
Date: 2025-01-21, Quantity: 20
Date: 2025-01-22, Quantity: 20
Date: 2025-01-23, Quantity: 20
Date: 2025-01-24, Quantity: 20
Date: 2025-01-25, Quantity: 20
Date: 2025-01-26, Quantity: 20
Date: 2025-01-27, Quantity: 20
```

4. **Transaction:**

```
Menu:
1. Check stock
2. Start a new transaction
3. Logout
4. Return transformer
5. Switch user
Enter your choice: 2
Enter your municipality: Maribojoc
Enter the transaction date (YYYY-MM-DD): 2025-01-01
Enter number of transformers: 20
Enter number of days for rental: 2
Enter applicant's name: Marc Alita
Enter venue: Bohol Gym
Enter approver's name: Ken Alita

How would you like to process the receipt?
1. Save to file
2. Display to screen
3. Save, Display and Print
```

5. **Return Transformer:**

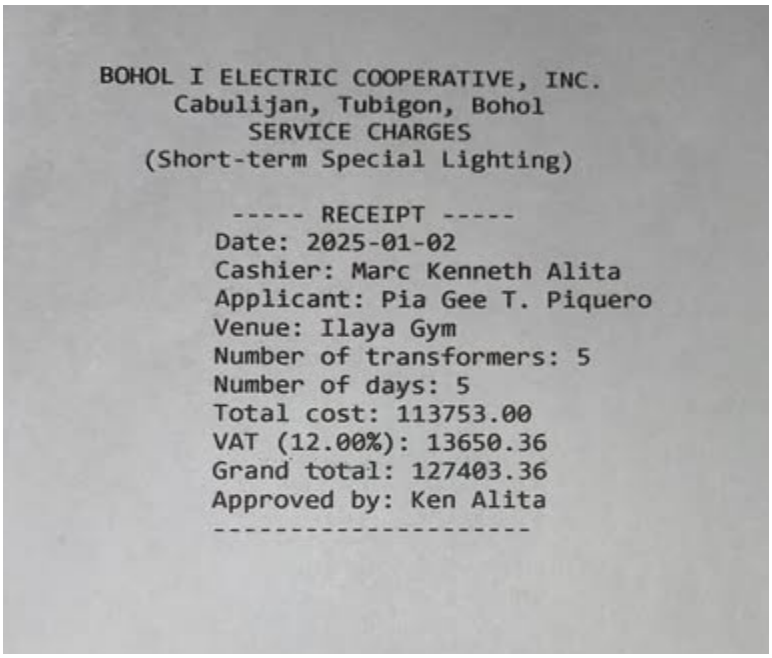
```
Menu:
1. Check stock
2. Start a new transaction
3. Logout
4. Return transformer
5. Switch user
Enter your choice: 4
Enter the transaction date (YYYY-MM-DD) for the return: 2025-01-01
Enter the number of transformers to return: 30
Successfully returned 30 transformers to stock.
```

6. **Receipt Generation:**

```
1. Save to file
2. Display to screen
3. Save, Display and Print
3
Receipt successfully saved to file: receipt_20250111_201138.txt
       BOHOL I ELECTRIC COOPERATIVE, INC.
            Cabulijan, Tubigon, Bohol
                 SERVICE CHARGES
          (Short-term Special Lighting)

               ----- RECEIPT -----
              Date: 2025-01-03
              Cashier: Mika Butchayo
              Applicant: Earl
              Venue: Ub Gym
              Number of transformers: 20
              Number of days: 2
              Total cost: 182004.80
              VAT (12.00%): 21840.58
              Grand total: 203845.38
              Approved by: Ken
              ---------------------
```

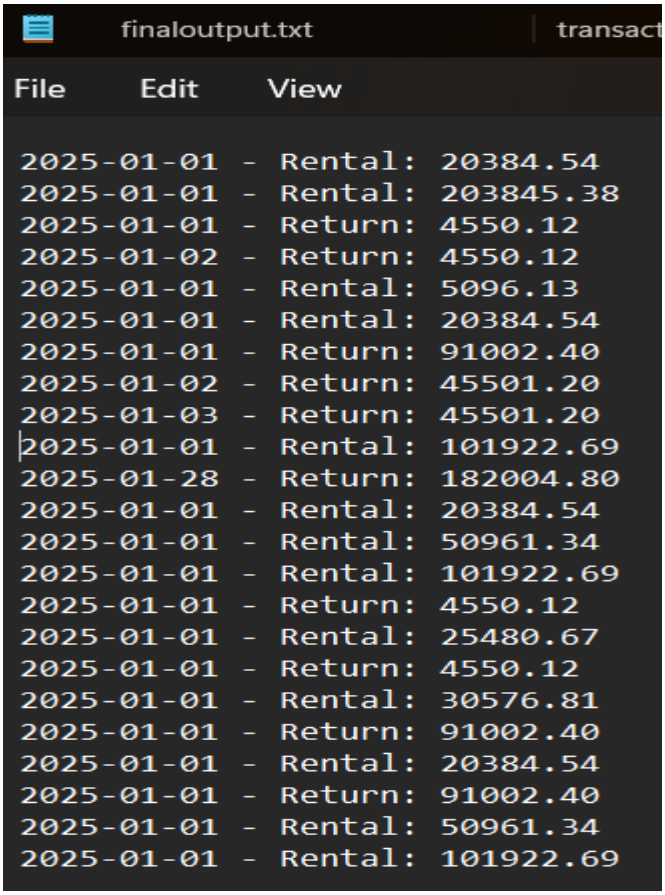
Printed output



```
BOHOL I ELECTRIC COOPERATIVE, INC.
Cabulijan, Tubigon, Bohol
SERVICE CHARGES
(Short-term Special Lighting)

----- RECEIPT -----
Date: 2025-01-02
Cashier: Marc Kenneth Alita
Applicant: Pia Gee T. Piquero
Venue: Ilaya Gym
Number of transformers: 5
Number of days: 5
Total cost: 113753.00
VAT (12.00%): 13650.36
Grand total: 127403.36
Approved by: Ken Alita
---------------------
```

7. **Rental and Retrun money**

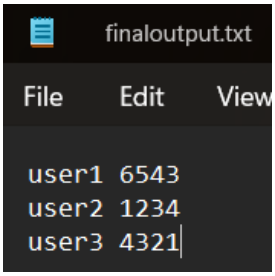


finaloutput.txt | transact

File Edit View

```
2025-01-01 - Rental: 20384.54
2025-01-01 - Rental: 203845.38
2025-01-01 - Return: 4550.12
2025-01-02 - Return: 4550.12
2025-01-01 - Rental: 5096.13
2025-01-01 - Rental: 20384.54
2025-01-01 - Return: 91002.40
2025-01-02 - Return: 45501.20
2025-01-03 - Return: 45501.20
2025-01-01 - Rental: 101922.69
2025-01-28 - Return: 182004.80
2025-01-01 - Rental: 20384.54
2025-01-01 - Rental: 50961.34
2025-01-01 - Rental: 101922.69
2025-01-01 - Return: 4550.12
2025-01-01 - Rental: 25480.67
2025-01-01 - Return: 4550.12
2025-01-01 - Rental: 30576.81
2025-01-01 - Return: 91002.40
2025-01-01 - Rental: 20384.54
2025-01-01 - Return: 91002.40
2025-01-01 - Rental: 50961.34
2025-01-01 - Rental: 101922.69
```

8. **Users**

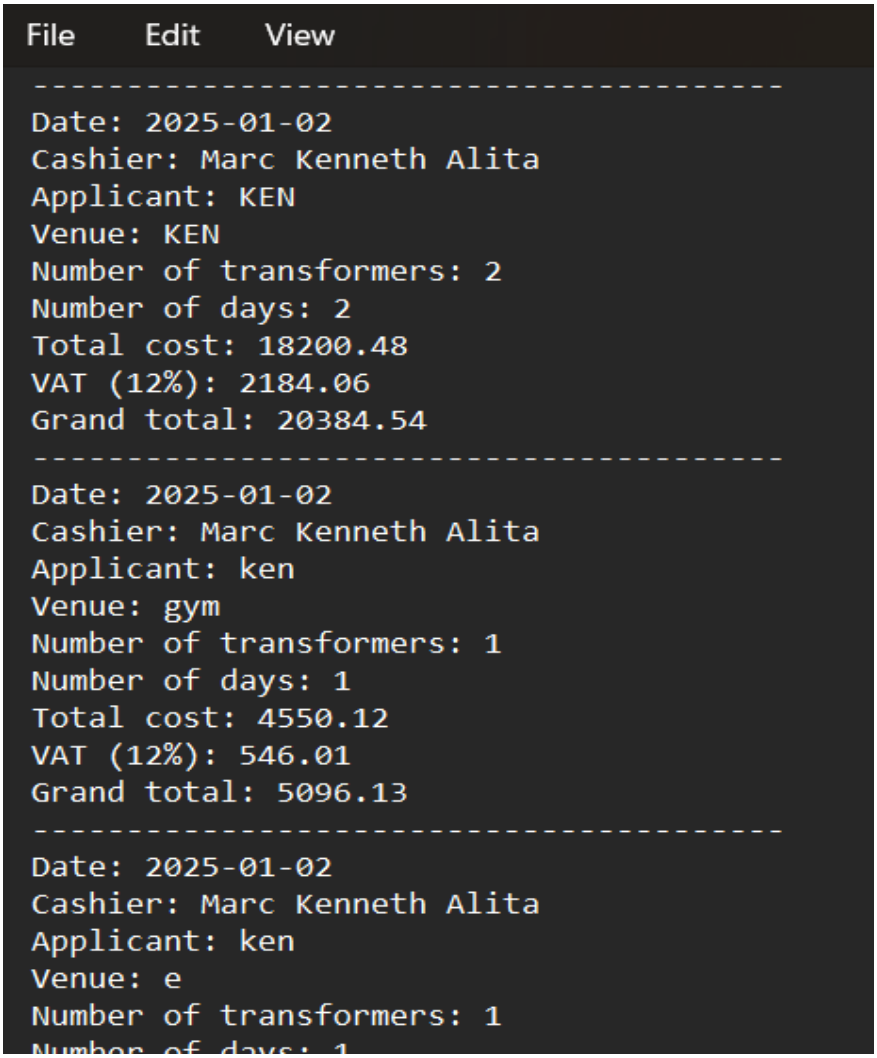


User 1 Marc Alita

User 2 Mika Butchaya

User 3 Nynl Ursos

9. **Transaction Log**



The Team

Name and Their Task:

- **Marc Kenneth A. Alita:** Project Manager, Documentation, Lead Developer, Backend Implementation, UI Design, Testing.

Member Curriculum Vitae with Pictures:



MARC KENNETH ALITA

BSCOMPE - 1

PERSONAL DATA

EMAIL:	mkaalita@universityofbohol.edu.ph
AGE:	19
SEX:	MALE
RELIGION:	ROMAN CATHOLIC
DATE OF BIRTH:	OCTOBER 23, 2005
CITIZENSHIP:	FILIPINO

EDUCATION

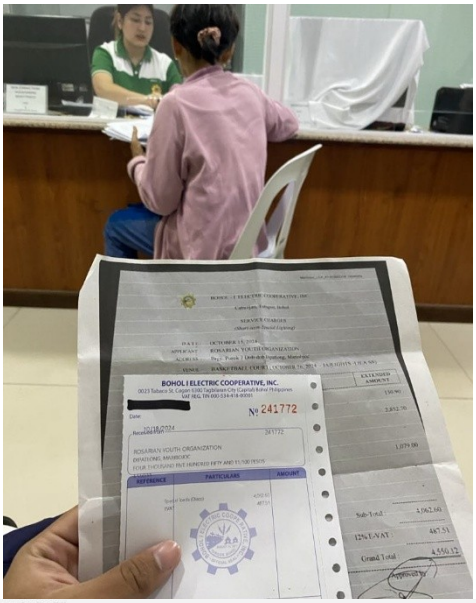
ELEMENTARY:	SANTA CRUZ ELEMENTARY SCHOOL
HIGH SCHOOL:	HOLY SPIRIT SCHOOL OG TAGBILARAN CITY
SENIOR HIGH SCHOOL:	HOLY SPIRIT SCHOOL OF TAGBILARAN CITY

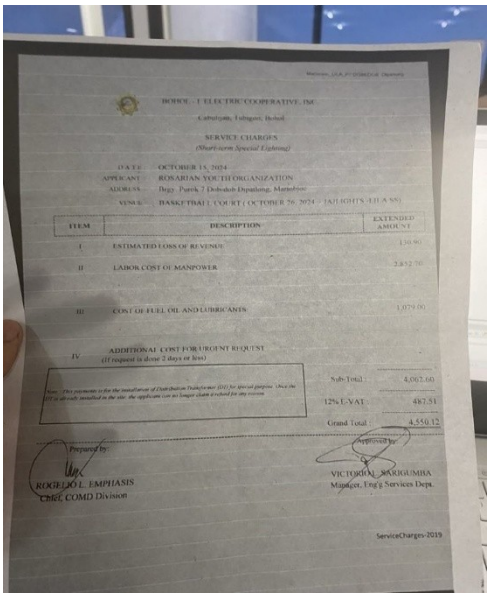
SKILLS

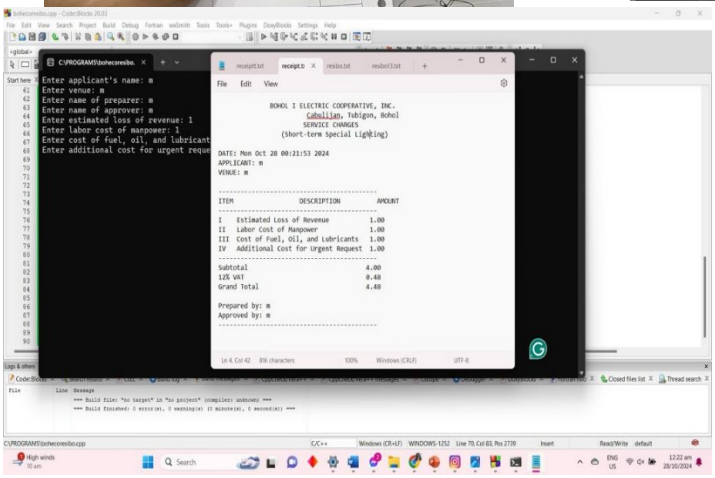
- CRITICAL THINKING
- PROBLEM SOLVING

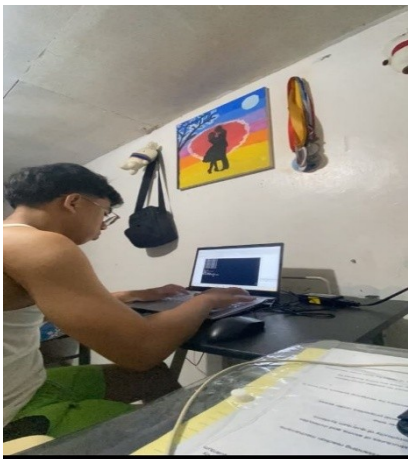
Photos

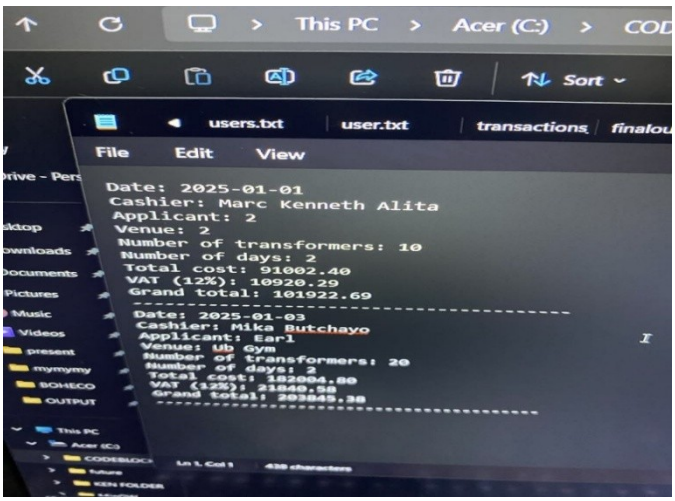
Documentation of the team during project development:

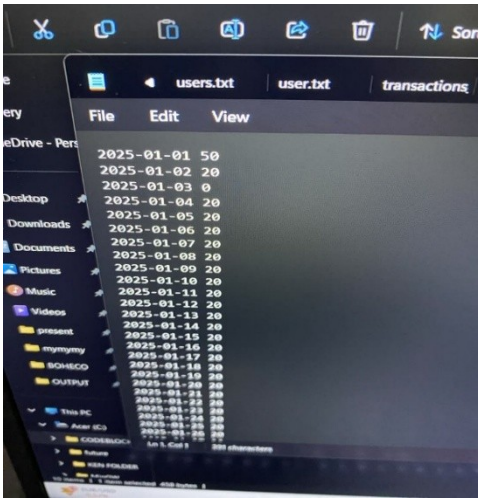


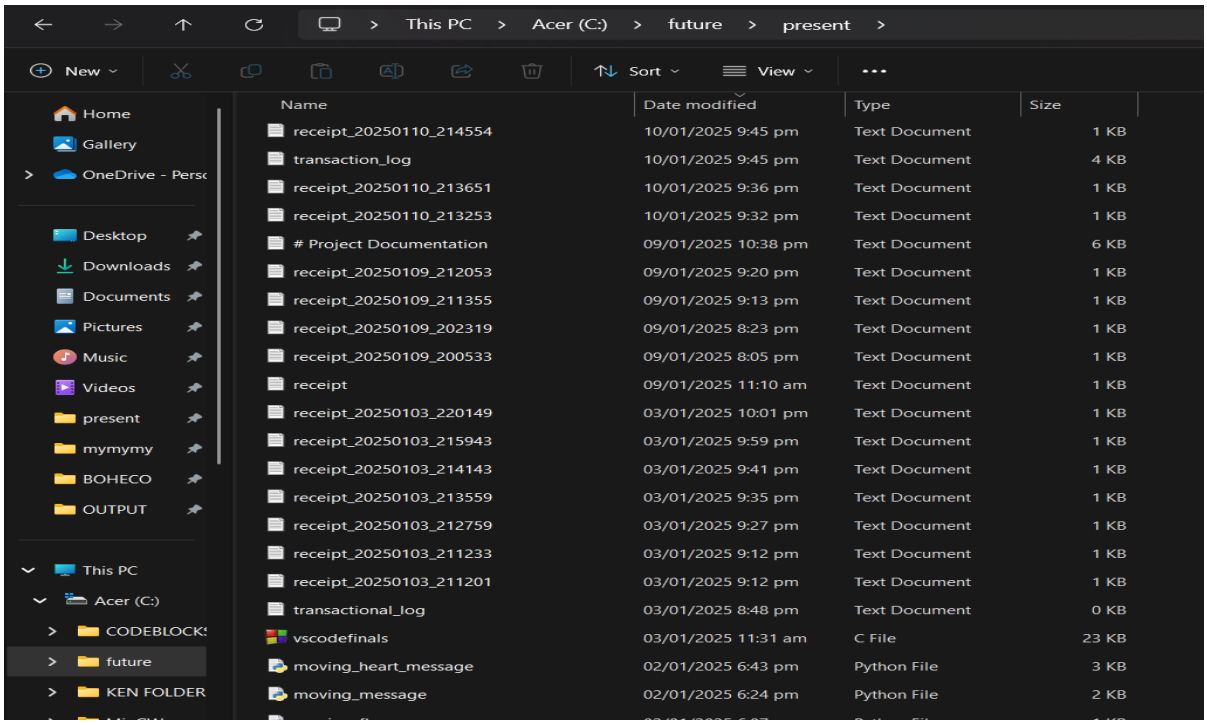




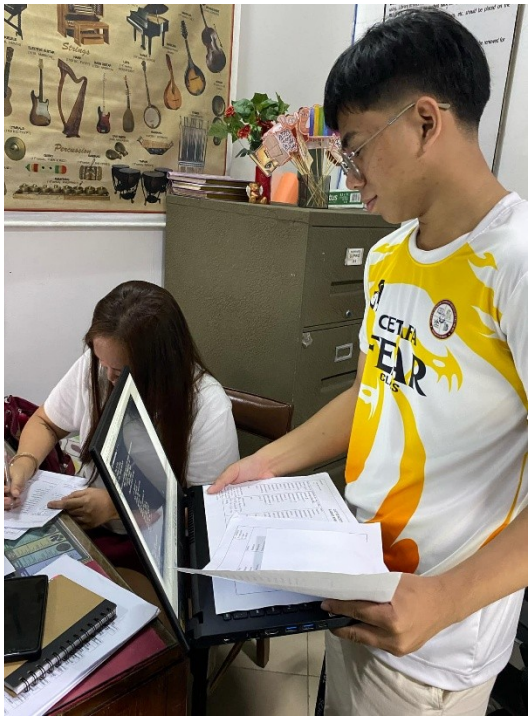







This PC > Acer (C:) > future > present
New
Sort
View
Home
Gallery
OneDrive - Perso
Desktop
Downloads
Documents
Pictures
Music
Videos
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mymymy
BOHECO
OUTPUT
This PC
Acer (C:)
CODEBLOCKS
future
KEN FOLDER
Name
Date modified
Type
Size
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transaction_log 10/01/2025 9:45 pm Text Document 4 KB
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Project Documentation 09/01/2025 10:38 pm Text Document 6 KB
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receipt 09/01/2025 11:10 am Text Document 1 KB
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receipt_20250103_215943 03/01/2025 9:59 pm Text Document 1 KB
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receipt_20250103_211201 03/01/2025 9:12 pm Text Document 1 KB
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vscodefinals 03/01/2025 11:31 am C File 23 KB
moving_heart_message 02/01/2025 6:43 pm Python File 3 KB
moving_message 02/01/2025 6:24 pm Python File 2 KB





Future Development

Recommendations for Future Development:

- **Graphical User Interface (GUI):** Develop a GUI for better usability and wider user adoption.
- **Mobile Application:** Create a mobile app version to allow for remote management and rental processing.
- **Cloud Storage Integration:** Integrate cloud storage for centralized and secure data management.
- **Expanded Equipment Rentals:** Extend the system to handle other equipment types, not just transformers.

Project Cost:

Approximately

- Development ₱10,000
- Software/Tools ₱25,00

Total: ₱35,000

Glossary:

- **MAX_USERS**: Maximum number of users the system can handle.
- **BASE_COST**: Base rental cost per transformer per day.
- **VAT**: Value Added Tax rate (12%).
- **FILENAME**: Name of the file used to store receipts.
- **STOCK_FILE**: File containing stock information.
- **LOG_FILE**: File used to log transaction details.
- **ADMIN_USERNAME & ADMIN_PASSWORD**: Hardcoded admin credentials.
- **cashier_names**: Predefined array mapping usernames to cashier names.
- **municipalities**: Array of valid municipalities where services are offered.
- **users**: Array of User structures containing loaded user data.
- **stock**: Array of Stock structures representing current stock levels.

Bibliography

- Official C Language Documentation: https://en.cppreference.com/
- GetDefaultPrinter Documentation: https://learn.microsoft.com/en-us/windows/win32/printdocs/getdefaultprinter
- GitHub Repositories: Stock management systems. https://github.com/
- https://www.w3schools.com/